Name: Sly, Noah, Carson **Date:** February 5th, 2026 **Class:** Engineering 4USA

MyDesign Portfolio

Element C Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design requirements, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design requirements, be sure to delete the instructions. Make sure to list your design requirements in ranked order, with design requirement #1 having the highest priority. If possible, list 5 requirements that are labeled as HIGH priority and 1-5 additional requirements that are labeled as LOW priority.

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1

Temperature Control (High)

The enclosure must keep the inside temperature between 20°C and 27°C (68°F–80°F) while printing, and it must never go above 30°C (86°F).

Research shows that when the outside temperature gets close to or above 30°C, filament can soften too early. This can cause clogs, weak prints, or messy layers. If the printer gets too hot, the cooling fans cannot work properly, and print quality goes down.

The printer's power brick must stay outside of the cardboard enclosure because it needs airflow. It should sit on tile or metal so heat does not warm up the cardboard.

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Stakeholder Feedback:
I talked to our technology teacher who uses the 3D printers regularly. He said that once the room gets close to 28–30°C, print quality starts to drop. He agreed that setting a specific maximum temperature makes this requirement clearer and easier to measure.

Design Requirement #2

Noise Reduction (High)

The enclosure must reduce the printer's sound by at least 10 decibels. Since most 3D printers run between 40–60 dB, the goal is to lower it to around 30–50 dB.

Sound escapes through gaps, especially corners and door openings. The box must be sealed tightly using tape or weather stripping. The vertical corner seams should be taped first because they leak the most sound.

It is not realistic to make a cardboard box completely silent, but it should noticeably reduce noise.

Stakeholder Feedback:
They said total silence is impossible with cardboard, but sealing corners properly makes a big difference. He recommended double-taping vertical seams.

Design Requirement #3

Fire Safety(High)

Because cardboard is flammable, the enclosure must include a battery-powered smoke detector either inside or directly above it.

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DIY enclosure safety guides warn that overheating electronics can become a fire risk. A smoke detector provides early warning if something goes wrong.

All main electronics, including the power supply, must remain outside the enclosure.

Stakeholder Feedback:
The teacher said that if cardboard is being used around electronics, a smoke detector is necessary in a classroom setting.

Design Requirement #4

Simple and Fast Access(High)

Students must be able to open the enclosure in under 5 seconds without tools or peeling tape.

If a print fails, it needs to be stopped quickly to prevent filament buildup or overheating. The door should use binder clips, magnets, or weights instead of tape so it can open immediately.

Stakeholder Feedback:
The teacher mentioned that students often panic during failed prints and struggle with tape. He suggested clips or magnets for faster access.

Design Requirement #5

Proper Airflow Ventilation(High)

The enclosure must include at least four small vent holes (1–2 cm wide) placed on the lower side panels.

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The motor and power components still need airflow to prevent overheating. The holes should not be placed on the top because dust in the classroom could fall inside. The vents should be covered with mesh or window screen to block dirt.

Stakeholder Feedback:
The teacher said top holes would collect dust. He recommended side vents with mesh covering.

Design Requirement #6

Structural Stability

The enclosure must stay stable and not collapse during long prints (2+ hours).

Corners must be reinforced with extra cardboard layers and strong tape. The box should not shift or bend while the printer is running.

Stakeholder Feedback:
The teacher said past student enclosures sometimes collapsed at the corners. Reinforcing edges is important for safety.

Design Requirement #7

Durable Observation Window

The enclosure must include a clear viewing window made from strong recycled plastic (such as clamshell packaging).

Students need to monitor prints to prevent “spaghetti” failures. Plastic wrap is too weak and can tear. The plastic should be firmly taped around all edges.

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Stakeholder Feedback:
The teacher said plastic wrap would not survive in a busy classroom. He suggested thicker recycled packaging plastic.

Design Requirement #8

Cable Control and Protection

The power cord must exit through a reinforced hole lined with double-layered tape so the cardboard does not rub against it.

The cord should also be secured to the outside of the box so if someone trips over it, the printer will not get pulled.

Stakeholder Feedback:
The teacher warned that cardboard edges can slowly wear down wires, so reinforcing the exit hole is important.

Design Requirement #9

Vibration Reduction

Soft materials such as cardboard stacks, felt pads, or old kitchen sponges should be attached to the bottom corners to absorb vibration.

A lot of printer noise comes from shaking the table. Dampening the base reduces the “thumping” sound.

Stakeholder Feedback:
The teacher suggested using recycled materials instead of buying squash balls to save money.

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